User Study

Background: What this study is about
This study evaluates an **AI-assisted maintenance question-answering system** designed to support engineers working with industrial assets. The goal is **not** to test language quality, but to assess whether the system's questions and answers are **operationally valid, data-grounded, and safe** for use in maintenance decision support. The system operates over **episode-level maintenance data**: each question refers to a *specific asset*, a *specific time window*, and a *fixed set of extracted facts* derived from raw telemetry and logs. All answers are expected to be verifiable against the underlying data. You are **not evaluating whether the answer is "smart" or "well written."** You are evaluating whether the question or answer:

- can be justified using the available data,
- stays within appropriate factual scope,
- avoids unsupported assumptions or overclaims,
- and aligns with how maintenance decisions should be reasoned about in practice.

Dataset used in this study (Microsoft Azure PdM)
All examples in this survey are derived from the **Microsoft Azure Predictive Maintenance (PdM) dataset[1]**, a publicly available industrial maintenance dataset. The dataset contains:

- **Hourly telemetry** from 100 machines
- (voltage, rotation speed, pressure, vibration)
- **Failure records** indicating component replacements after breakdowns
- **Non-fatal error events** that do not cause shutdowns
- **Maintenance logs** (scheduled and unscheduled)
- **Machine metadata** (model type, age)

From this raw data, we construct **fixed 24-hour episodes** for each machine. For each episode, we extract:

- summary statistics and trends of sensor signals,
- counts of error events and distinct error types,
- time since last maintenance for relevant components,
- and the exact time window and data provenance.

Each question you see refers **only** to this extracted information. No external knowledge, hindsight, or additional assumptions are allowed. What you are being asked to judge You will see different **types of questions**, reflecting real maintenance workflows:

- **Temporal / counting** (e.g., how many errors occurred)
- **Descriptive** (e.g., average sensor values)
- **Diagnostic** (evidence supporting a labeled fault)
- **Counterfactual** (what would change under a hypothetical intervention)
- **Action-oriented** (whether available evidence is sufficient to justify action)

For each item, you are asked to judge **properties of the question itself** (and sometimes the answer), such as:

- Is it answerable from the data?
- Is it clearly grounded in observable evidence?
- Is it ambiguous to a human expert?
- Does it overstep into causality, intent, or prescription without support?

There are **no trick questions**. Disagreement between experts is expected and valuable. Important clarification This is **not** a test of whether the AI's recommendation is "correct."

It is a test of whether the **questioning and reasoning structure** respects real-world maintenance constraints. If something feels unsafe, underspecified, misleading, or unjustified by the data, you should mark it accordingly even if the answer sounds plausible.

Required

Temporal questions

Localize events in time or aggregate occurrences (e.g., onset time, error counts)

1. 📘 Episode Summary (Human-Readable)
Asset: machine_82 (compressor)
Equipment type: Electric motor driven rotating machine
Time window: 2015-01-27 06:00 → 2015-01-28 06:00 (24 hours)
Telemetry samples: 24
Error count in window: 1
Distinct error types: 1
Episode ID (fact_id): pdm_m82_comp1_2015-01-28T06
Short interpretation: during the 24-hour episode for compressor machine_82 there was one error event and one distinct error type recorded.

❓ **Question Being Evaluated**
Between 2015-01-27 06:00:00 and 2015-01-28 06:00:00 for asset machine_82, how many distinct error types occurred?

🤖 System Output (from the JSON)
Direct Answer: There were **1** distinct error types in this window.
Reasoning Answer: In this episode the feature distinct_error_types_last_window is **1**, and the total error count is **1**. Telemetry points in window: **24**.

📊 Key Extracted Evidence

- error_count_last_window = **1**
- distinct_error_types_last_window = **1**
- telemetry_points_in_window = **24**
- Source file: PdM_telemetry.csv (row: **80**)
- fact_id: pdm_m82_comp1_2015-01-28T06

📝 Your Evaluation (Quick — read summary first, then answer)
You are judging the **question** (is it data-verifiable, clear, scoped), not how well it's written.

Q1. Answerability (Binary)
Can this question be answered using only the provided extracted facts and the PdM dataset (no external assumptions)?

Guideline:
Yes = can verify from distinct_error_types_last_window, error_count_last_window, telemetry timestamps.
No = needs causality, prediction, intent, or outside data.

◯ Yes

◯ No

2. **Q2. Data Grounding (1–5)**
How well is this question grounded in observable data (events, telemetry, timestamps)?

	Not grounded	Weakly grounded	Moderately grounded	Strongly grounded	Fully grounded
How well is this question grounded in observable data (events, telemetry, timestamps)?	◯	◯	◯	◯	◯

3. **Q3. Ambiguity (1–5)**
How unambiguous is this question given the provided data? *

	Extremely ambiguous	Ambiguous	Somewhat clear	Mostly clear	Completely unambiguous
How unambiguous is this question given the data?	◯	◯	◯	◯	◯

4. **Q4. Scope Correctness (Binary)**
Does this question stay strictly within factual verification (no explanation, causality, or decision-making)?

Guideline:
Yes → pure counting / timestamp queries.
No → asks "why" / "what to do" / or "what would happen if…" *

◯ Yes (fact verification only)

◯ No (requires reasoning beyond facts)

5. **Q5. Overclaim Detection**

Does this question implicitly assume claims not supported by the data (e.g., causality, necessity)?

Guideline:
Overclaim = the question presupposes cause/effect or prescriptive conclusions that are not derivable from the listed features. *

◯ No

◯ Yes

Descriptive questions

Report observed values or summary statistics (e.g., mean vibration)

6. 📘 Episode Summary (Human-Readable)
Asset: machine_99 (model1)
Equipment category: Rotating equipment
Equipment type: Electric motor driven rotating machine
Time window: 2015-01-01 03:00 → 2015-01-02 03:00 (24 hours)
Telemetry samples: 22
Extracted feature used: vibration_mean
Episode ID (fact_id): pdm_m99_comp3_2015-01-02T03
Short interpretation: During this 24-hour episode, the system computed the average vibration level from 22 telemetry readings.

❓ **Question Being Evaluated**
During the time window 2015-01-01 03:00:00 to 2015-01-02 03:00:00 for asset machine_99, what was the average vibration level?
(Vibration sensors monitor machinery for abnormal vibrations, which can indicate issues like misalignment, imbalance, or wear.)

🤖 System Output (from the JSON)
Direct Answer: The average vibration level was approximately **40.15**.
Reasoning Answer: In this episode for asset machine_99, the feature vibration_mean is **40.15**, computed over **22 telemetry points** in the window 2015-01-01 03:00:00 to 2015-01-02 03:00:00.

📊 Key Extracted Evidence

- vibration_mean = **40.1486**
- telemetry_points_in_window = **22**
- Source file: PdM_telemetry.csv (row: **1**)
- fact_id: pdm_m99_comp3_2015-01-02T03

Sensor description provided: Vibration sensors monitor machinery for abnormal vibrations, which can indicate issues like misalignment, imbalance, or wear.
📝 Your Evaluation: You are evaluating the **question formulation**, not whether 40.15 is numerically correct.

Q1. Answerability (Binary)
Can this question be answered using only the provided extracted facts and the PdM dataset (no external assumptions)?

Guideline:
Yes = The answer is directly computable from listed feature values.
No = Requires causal reasoning or interpretation beyond provided features.

◯ Yes

◯ No

7. **Q2. Data Grounding (1–5)**
How well is this question grounded in observable data (events, telemetry values, timestamps)?

	Not grounded	Weakly grounded	Moderately grounded	Strongly grounded	Fully grounded
How well is this question grounded in observable data (events, telemetry, timestamps)?	◯	◯	◯	◯	◯

8. **Q3. Ambiguity (1–5)**
How unambiguous is this question given the provided data? *

	Extremely ambiguous	Ambiguous	Somewhat clear	Mostly clear	Completely unambiguous
How unambiguous is this question given the data?	◯	◯	◯	◯	◯

9. **Q4. Scope Correctness (Binary)**
Does this question stay strictly within factual verification (no explanation, causality, or decision-making)?

Guideline:
Yes → Pure retrieval of computed feature value.
No → Implies diagnosis, cause, or recommendation. *

◯ Yes (fact verification only)

◯ No (requires reasoning beyond facts)

10. **Q5. Overclaim Detection**

Does this question implicitly assume claims not supported by the data (e.g., causality, necessity)?

Guideline:
Overclaim = presupposing mechanical failure, causality, or necessity from vibration value alone. *

◯ No

◯ Yes

Diagnostic questions

Explain why an episode has a given label by identifying and grounding the salient evidence.

11. 📘 Episode Summary (Human-Readable)
Asset: machine_45 (model3)
Equipment category: Rotating equipment
Equipment type: Electric motor driven rotating machine
Time window: 2015-01-01 03:00 → 2015-01-02 03:00 (24 hours)
Telemetry samples: 22
Assigned failure label: comp1
Failure description: Motor stator / power circuit fault
Severity: High
Associated sensor: Voltage (volt)
Episode ID (fact_id): pdm_m45_comp1_2015-01-02T03

📊 Observed Feature Values in This Episode

- volt_mean = **190.520**
- volt_max = **214.528**
- volt_trend = **0.255**

(These are the extracted values used for labeling.)

❓ **Question Being Evaluated**
Why is this diagnostic episode for asset machine_45 labeled "comp1" (Motor stator / power circuit fault) over the time window 2015-01-01 03:00:00 to 2015-01-02 03:00:00?

🤖 System Output (from JSON)
Direct Answer: This episode is labeled "comp1" because the observed features match the typical indicators associated with this failure mode.
Reasoning Answer: Key observed features:

- volt_mean = 190.520
- volt_max = 214.528
- volt_trend = 0.255

Typical indicators include:

- Elevated voltage mean
- Positive or unstable voltage trend
- Rotational speed near normal

Suggested inspection steps:

- Inspect power supply quality and protection settings
- Inspect stator windings and insulation condition
- Verify cabling, terminations, and grounding

Severity rating: High.

🧠 What You Are Evaluating
You are judging whether the **question itself** asks for:

- Evidence-based justification
  OR
- A causal explanation of the failure

This distinction is important.

📝 Your Evaluation

Q1. Evidence Supportability (Binary)
Does this question ask for observable evidence supporting the assigned label, rather than asking for a causal explanation?

Guideline
✔ **Yes** → The question asks the user to identify which observed features justify the label.
✘ **No** → The question asks for mechanical cause, root cause, or physical failure mechanism explanation.

Clarification

- Acceptable (evidence-focused): "Which observed features support labeling this episode as comp1?"
- Causal drift (not evidence-focused): "What caused this motor stator failure?"

Your task is to determine which category this question belongs to.

- Yes
- No

Counterfactual questions

Estimate how a specified intervention would change failure risk (direction and magnitude) using the risk simulator.

12. 📘 Episode Summary (Human-Readable)

Asset: machine_87 (model2)
Equipment category: Rotating equipment
Equipment type: Electric motor driven rotating machine
Time window: 2015-01-01 03:00 → 2015-01-02 03:00 (24 hours)
Telemetry samples: 22
Observed error events: 1
Assigned failure label: comp1
Failure description: Motor stator / power circuit fault
Severity: High
Maintenance recency feature:

- hours_since_last_maint_comp1 = **477.0 hours**

Episode ID (fact_id): pdm_m87_comp1_2015-01-02T03

🔄 Hypothetical Intervention
Assume maintenance targeting comp1 had been performed immediately before the window. This would reset: hours_since_last_maint_comp1 from **477.0 → 0.0** No other features are changed.

❓ **Question Being Evaluated**
If maintenance targeting comp1 had been performed immediately before the window (resetting hours_since_last_maint_comp1 from 477.0 to 0), how would the risk of failure change?

🤖 System Output (from JSON)
Direct Answer: The risk of failure would **decrease**.
Reasoning Answer (model-based): Baseline risk:
P(failure) ≈ **1.000** Post-intervention risk:
P(failure) ≈ **0.000** Δ risk ≈ **−1.000** 📊 Counterfactual Model Outputs

- risk_before = **1.0**
- risk_after ≈ **0.0000107**
- delta_risk ≈ **−0.999989**
- Direction: **decrease**

Intervention applied: do(hours_since_last_maint_comp1 = 0.0)

🧠 What You Are Evaluating
You are not judging whether the simulator is correct. You are judging whether the **question formulation**:

- Stays within a clearly defined hypothetical intervention
- Or makes uncontrolled real-world causal claims

📝 Your Evaluation
Q1. Counterfactual Dependency Validity (Binary)
Does this question restrict itself to a clearly defined hypothetical feature change, without asserting real-world causal correctness?

Guideline
✔ **Yes** → The question clearly specifies:

- The exact feature being changed
- The new value
- That the scenario is hypothetical
- That risk change is computed by a model

✘ **No** →
The question:

- Implies real-world causality beyond the stated feature
- Assumes physical correctness
- Makes claims about actual failure mechanism rather than model-based risk

* ◯ Yes

◯ No

Action-recommendation questions

Recommend whether to act (e.g., create a work order) by combining grounded evidence with counterfactual risk and confidence constraints.

13. 📘 Episode Summary (Human-Readable)
Asset: machine_86 (model3)
Equipment category: Rotating equipment
Equipment type: Electric motor driven rotating machine
Time window: 2015-01-01 03:00 → 2015-01-02 03:00 (24 hours)
Telemetry samples: 22
Observed error events: 0
Failure label: comp1
Failure severity: High
Model-estimated risk of failure:≈ **1.00** (threshold = 0.50)
Most informative sensor: Voltage (volt)
Episode ID (fact_id): pdm_m86_comp1_2015-01-02T03

📊 Extracted Features Used (Subset — key decision-relevant signals)

- Voltage statistics: volt_mean, volt_std, volt_max, volt_trend
- Rotation statistics
- Pressure statistics
- Vibration statistics
- hours_since_last_maint_comp1
- error_count_last_window
- Machine age and model indicators

❓ **Question Being Evaluated**
For rotating_equipment machine_86 in the time window 2015-01-01 03:00:00 to 2015-01-02 03:00:00, should a maintenance work order be opened now, or is it acceptable to continue monitoring?

🤖 System Output (from JSON)
Direct Answer: A maintenance work order should be opened now.
Reasoning Answer:

- Predicted probability of failure ≈ **1.00**
- Failure severity: **High**
- Associated failure mode: comp1
- Recommended inspection steps include:
  -Inspect power supply quality and protection settings
  -Inspect stator windings and insulation
  -Verify cabling and grounding 📈 Model Risk Details

- risk ≈ **0.928**
- Highest predicted class: comp1
- Healthy probability ≈ extremely low

🧠 What You Are Evaluating
You are **not** judging whether the maintenance decision is correct. You are judging whether the **question formulation**:

- Evaluates whether available evidence is sufficient
  OR
- Presupposes that action is required

📝 Your Evaluation
Q1. Actionability Sufficiency (Binary)
Does this question evaluate whether the available extracted facts justify a maintenance decision, rather than assuming the decision is already correct?

Guideline
✔ **Yes** →The question asks whether evidence supports a decision (evidence sufficiency framing).
✘ **No** →The question presumes the action should occur and merely asks which action to take. *

◯ Yes

◯ No